

Design Setup (Input)

- Dropout Rate: 10%
- Recruitment Rate: 15 patients / month
- Randomization Block Size: 6
- Control Median PFS: 8 months
- Number of Treatments 4

Drop-the-losers Design: Phase II

- Stage 1: 50 events
- Stage 2: 70 events
- Stage 3: 90 events

Drop-the-losers Design: Phase III

- Stage 1: 100 events
- Stage 2: 200 events
- Stage 3: 400 events

Parallel Simulation Loop:

- Simulations run in parallel using the package *foreach* loop
- Simulates recruitment, randomization, and endpoint generation
- Censoring

Futility Analysis:

- Calculates HR, test statistic, p-value for each treatment arm.

Simulation Setup:

- Number of Simulations: 200'000 and 1'000'000
- 1. Null Hypothesis: No effect
- 2. Hybrid 1: One effective drug
- 3. Hybrid 2: Two effective drugs
- 4. Hybrid 1 + 1: One effective, one half-effective
- 5. Harmful Treatment
- 6. All Effective: All drugs show an effect.

```
# Endpoint simulation (PFS)
PFS <- ifelse(trt == 0, rexp(length(recruit), log(2) / medianPFS_control),
  rexp(n = length(recruit), as.numeric(log(2) / medianPFS_control * Results[i, trt + 2])))
```

$T_{control} \sim \text{Exp}(\lambda_{control}) = \text{Exp}\left(\frac{\log(2)}{M_{control}}\right)$

Cox-Regression Model:

$h(t|X) = h_0(t) * e^{(\beta_1 * X_1)}$

```
coxph(Surv(ad_PFS, event_PFS) ~ ad_trt
```

$T_{treatment} \sim \text{Exp}(\lambda_{treatment}) = \text{Exp}\left(\frac{\log(2)}{M_{control}}\right) \times HR$

Futility Analysis and or drop the worst test statistics

Log Rank Test:

$\chi^2 = \frac{(O_1 - E_1)^2}{V}$

```
surfdiff(Surv(ad_PFS, event_PFS) ~ ad_trt
```

Decision based on the HR or Test statistic

Test statistics from the Cox Regression Model and from the Log Rank Test

Scenario	TreatmentArm	Simulation	AnalysisType	Time_Analysis	Outcome	HR	TestStatistic_Cox	TestStatistic_LogRank	Dropped
1	1	1	Interim 1	13.45135	Continue	0.6755952	-1.04118907	1.096189457	False
1	2	1	Interim 1	11.44947	Dropped due to Futility	1.0300956	0.15468724	0.023934791	True
1	3	1	Interim 1	13.45135	Continue	0.9232147	-0.64618783	0.419414569	False
1	4	1	Interim 1	13.45135	Dropped due to Drop-the-Loser Design	0.948015	-0.57041656	0.326510463	True
1	1	1	Interim 2	18.41185	Continue	0.8130886	-0.70781675	0.502607675	False
1	3	1	Interim 2	18.40082	Dropped due to Futility	1.0070193	0.0736778	0.005428612	True
1	1	1	Final Analysis	28.38477	Continue	0.9612026	-0.19729584	0.038930653	False
2	1	1	Interim 1	13.85829	Dropped due to Futility	1.5986132	1.25352446	1.597766486	True
2	2	1	Interim 1	14.13932	Dropped due to Futility	1.1580248	0.78026845	0.612757382	True